

# Practical Demo: Hedging with Options on Treasury Futures

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## Learning Objectives

After completing this demo you should be able to read a Treasury futures option quote, separate intrinsic value from time value, calculate a Black futures-option benchmark, size a protective-put overlay with DV01, compare its scenario P&L with a linear futures hedge, and explain basis, volatility, and time-decay risks.

# 1 Inputs and Assumptions

All inputs are **illustrative, market-like inputs**, not live quotes. The option is treated as European for Black-model transparency; actual contract exercise and margin conventions must be checked. The bond portfolio is mapped to one Treasury futures contract with DV01; the CTD, conversion factor, basis, convexity, transaction costs, margin cash flows, and early exercise are held fixed or ignored.

| Workflow stage       | In this case                                                |
|----------------------|-------------------------------------------------------------|
| Observed or supplied | Futures, strike, quoted premium, expiry, and portfolio DV01 |
| Estimated            | Black benchmark value and futures-equivalent exposure       |
| Diagnosed            | Quote-versus-model gap, intrinsic value, and time value     |
| Used for             | Choosing between a linear hedge and asymmetric protection   |

```
import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

portfolio_value = 25_000_000
portfolio_dv01 = 18_000
futures_price = 112.00
strike = 111.00
futures_dv01 = 78.00
dollars_per_point = 1_000
time_to_expiry = 0.50
risk_free_rate = 0.042
volatility = 0.075
```

## 2 Treasury Option Quote Convention

Treasury futures and their options are quoted in price points and fractions. For a \$100,000 face-value contract, one full point is \$1,000 and 1/64 of a point is \$15.625. A premium of 1-54 means

$$1 + \frac{54}{64} = 1.84375 \text{ points,}$$

or \$1,843.75 per contract. Price points, dollars, and yield basis points are different units.

```
quoted_points = 1 + 54 / 64
quoted_dollars = quoted_points * dollars_per_point
pd.Series({"Quoted premium (points)": quoted_points,
          "Quoted premium ($/contract)": quoted_dollars})
```

```
Quoted premium (points)          1.8438
Quoted premium ($/contract)      1843.7500
dtype: float64
```

### 3 Intrinsic Value, Time Value, and Black's Model

For a futures put, intrinsic value is  $\max(K - F, 0)$ . Black's European futures-put formula is

$$p = e^{-rT} [KN(-d_2) - FN(-d_1)],$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

```
def normal_cdf(x):
    x = np.asarray(x, dtype=float)
    return 0.5 * (1 + np.vectorize(math.erf)(x / np.sqrt(2)))

def black_futures_put(F, K, T, r, sigma):
    vol_time = sigma * np.sqrt(T)
    d1 = (np.log(F / K) + 0.5 * sigma**2 * T) / vol_time
    d2 = d1 - vol_time
    return np.exp(-r * T) * (K * normal_cdf(-d2) - F * normal_cdf(-d1))

model_put_points = float(black_futures_put(futures_price, strike, time_to_expiry,
                                           risk_free_rate, volatility))
intrinsic_points = max(strike - futures_price, 0)
time_value_points = model_put_points - intrinsic_points
```

```
pd.Series({
    "Black put value (points)": model_put_points,
    "Black put value ($/contract)": model_put_points * dollars_per_point,
    "Intrinsic value (points)": intrinsic_points,
    "Time value (points)": time_value_points,
    "Illustrative quoted premium (points)": quoted_points,
}).round(4)
```

```
Black put value (points)          1.8531
Black put value ($/contract)      1853.0571
Intrinsic value (points)          0.0000
Time value (points)              1.8531
Illustrative quoted premium (points) 1.8438
dtype: float64
```

The quoted premium and model value need not match. A trader can interpret their difference as a signal to investigate volatility, exercise style, liquidity, model inputs, and bid-ask effects—not as an automatic arbitrage.

## 4 Size the Protective Put Overlay

First map the portfolio into futures-equivalent contracts:

$$N_{fut-eq} = \frac{DV01_{portfolio}}{DV01_{futures}}.$$

For a one-for-one static floor at expiration, buy approximately one put per futures-equivalent contract. A current delta hedge would generally require a different count and rebalancing.

```
exact_futures_equivalents = portfolio_dv01 / futures_dv01
contracts = int(np rint(exact_futures_equivalents))
premium_cost = contracts * quoted_dollars
mapped_dv01 = contracts * futures_dv01
residual_dv01 = portfolio_dv01 - mapped_dv01

pd.Series({
    "Exact futures equivalents": exact_futures_equivalents,
    "Static puts purchased": contracts,
    "Premium cost ($)": premium_cost,
```

```
"Residual DV01 after rounding ($/bp)": residual_dv01,
}).round(2)
```

|                                      |           |
|--------------------------------------|-----------|
| Exact futures equivalents            | 230.77    |
| Static puts purchased                | 231.00    |
| Premium cost (\$)                    | 425906.25 |
| Residual DV01 after rounding (\$/bp) | -18.00    |

dtype: float64

## 5 Futures-Price and Yield Scenarios

Approximate the futures-price change from a Treasury yield shock:

$$\Delta F \approx -\frac{DV01_{futures}}{\$1,000 \text{ per point}} \Delta y_{bp}.$$

The mapped bond P&L moves like a long futures position. The protective put pays only below the strike. A short-futures hedge offsets mapped price changes linearly in both directions.

```
yield_shocks_bp = np.array([-75, -40, 0, 40, 75])
terminal_futures = futures_price - futures_dv01 / dollars_per_point * yield_shocks_bp
mapped_bond_pnl = contracts * dollars_per_point * (terminal_futures - futures_price)
put_payoff = contracts * dollars_per_point * np.maximum(strike - terminal_futures, 0)
protective_put_pnl = mapped_bond_pnl + put_payoff - premium_cost
short_futures_pnl = -mapped_bond_pnl
linear_hedge_pnl = mapped_bond_pnl + short_futures_pnl

scenario_table = pd.DataFrame({
    "Yield shock (bp)": yield_shocks_bp,
    "Terminal futures": terminal_futures,
    "Mapped bond P&L ($)": mapped_bond_pnl,
    "Put payoff ($)": put_payoff,
    "Bond + put P&L after premium ($)": protective_put_pnl,
    "Bond + short futures P&L ($)": linear_hedge_pnl,
})
scenario_table.round(0)
```

|   | Yield shock (bp) | Terminal futures | Mapped bond P&L (\$) | Put payoff (\$) | Bond + put P&L after pre |
|---|------------------|------------------|----------------------|-----------------|--------------------------|
| 0 | -75              | 118.0            | 1351350.0            | 0.0             | 925444.0                 |
| 1 | -40              | 115.0            | 720720.0             | 0.0             | 294814.0                 |
| 2 | 0                | 112.0            | 0.0                  | 0.0             | -425906.0                |
| 3 | 40               | 109.0            | -720720.0            | 489720.0        | -656906.0                |
| 4 | 75               | 106.0            | -1351350.0           | 1120350.0       | -656906.0                |

The put preserves gains when yields fall but requires an upfront premium. The linear futures hedge has no option premium in this simplified comparison, but it gives up gains as well as losses.

```

fine_shocks = np.linspace(-100, 100, 401)
fine_futures = futures_price - futures_dv01 / dollars_per_point * fine_shocks
fine_bond = contracts * dollars_per_point * (fine_futures - futures_price)
fine_put = fine_bond + contracts * dollars_per_point * np.maximum(strike - fine_futures, 0)
fine_linear = np.zeros_like(fine_shocks)

fig, ax = plt.subplots(figsize=(7.6, 4.2))
ax.plot(fine_shocks, fine_bond / 1_000, linewidth=2, label="Unhedged mapped bond")
ax.plot(fine_shocks, fine_put / 1_000, linewidth=2, label="Bond + protective puts")
ax.plot(fine_shocks, fine_linear, linestyle="--", linewidth=2, label="Bond + short futures")
ax.axhline(0, color="black", linewidth=0.8)
ax.set(xlabel="Treasury yield shock (bp)", ylabel="P&L ($ thousands)",
       title="Expiration P&L after option premium")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()

```

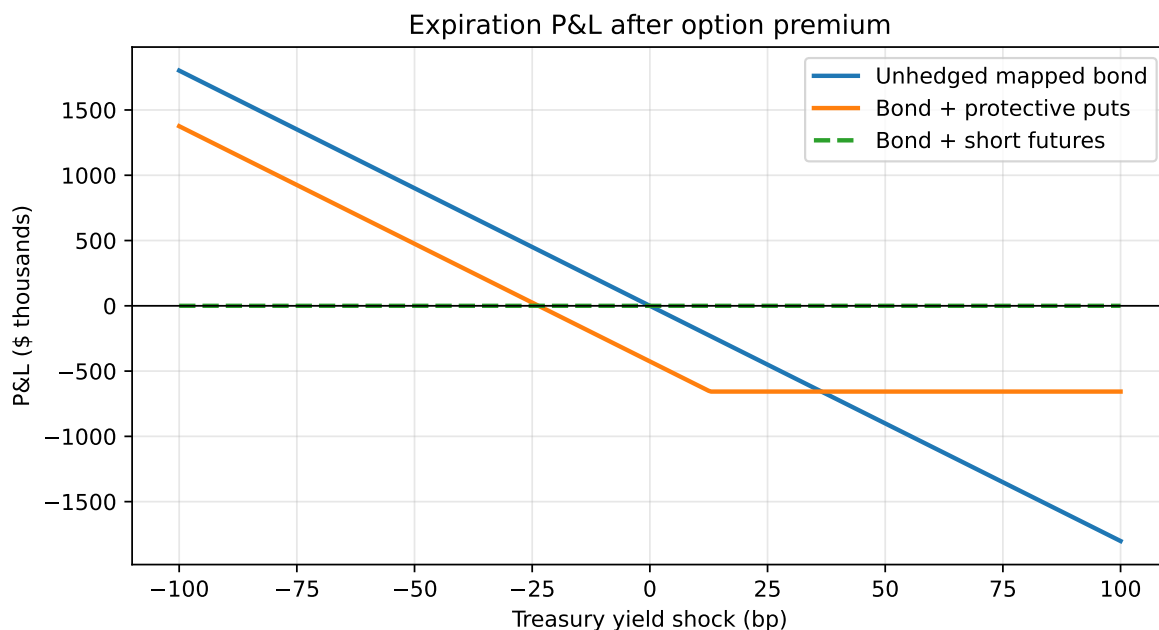


Figure 1: A protective put creates a floor; a short-futures hedge removes both upside and downside

## 6 Risks the Simple Comparison Omits

- **Basis risk:** the bond portfolio and Treasury futures may move differently.
- **Curve risk:** a single contract cannot hedge every maturity exposure.
- **Volatility risk:** option value changes when implied volatility changes.
- **Time decay:** the put loses time value as expiration approaches, all else equal.
- **Delta drift:** the option's sensitivity changes with price, time, and volatility.
- **Exercise and margin conventions:** listed contracts can differ from the European, premium-paid simplification.
- **Liquidity and roll risk:** bid-ask costs and hedge replacement affect realized results.

## 7 Validation Checks

```
assert model_put_points >= intrinsic_points >= 0
assert abs(quoted_dollars - 1_843.75) < 1e-10
assert abs(residual_dv01) <= 0.5 * futures_dv01 + 1e-10
assert np.allclose(linear_hedge_pnl, 0)
```

```

assert protective_put_pnl[-1] > mapped_bond_pnl[-1]

pd.Series({
    "Model put value exceeds intrinsic value": model_put_points >= intrinsic_points,
    "1-54 quote converts to $1,843.75": True,
    "Rounding residual is within half one contract DV01": True,
    "Linear futures hedge offsets mapped P&L": bool(np.allclose(linear_hedge_pnl, 0)),
    "Put improves the rising-yield outcome": protective_put_pnl[-1] > mapped_bond_pnl[-1],
})

```

|                                                    |      |
|----------------------------------------------------|------|
| Model put value exceeds intrinsic value            | True |
| 1-54 quote converts to \$1,843.75                  | True |
| Rounding residual is within half one contract DV01 | True |
| Linear futures hedge offsets mapped P&L            | True |
| Put improves the rising-yield outcome              | True |
| dtype: bool                                        |      |

## 8 Financial Interpretation

Contract selection begins with the risk, not the forecast. A short futures hedge is appropriate when the objective is to remove exposure linearly. A protective put is appropriate when the manager wants a loss floor while retaining favorable bond-price gains. That asymmetry is valuable, but its premium, volatility exposure, time decay, basis, and expiration must be budgeted explicitly.